Q - Longest Substring Without Repeating Characters

- **Find the length of the longest substring without repeating characters**

```
def longest_substring_without_repeating(s):
    seen = set()        # To store characters in the current window
    left = 0            # Left pointer of the sliding window
    max_len = 0         # Result variable
    for right in range(len(s)):
        while s[right] in seen:
            seen.remove(s[left])
            left += 1
        seen.add(s[right])
        max_len = max(max_len, right - left + 1)

    return max_len

# Driver Code
s = input().strip()  #abcabcbb
print(longest_substring_without_repeating(s))
```

Q – Compress a String

```
def compress_string(s):
    if not s:
        return ""
    compressed = ""
    count = 1
    for i in range(1, len(s)):
        if s[i] == s[i - 1]:
            count += 1
        else:
            compressed += s[i - 1] + str(count)
            count = 1
    # Add last character and its count
    compressed += s[-1] + str(count)
    # Return the smaller one (optional)
    return compressed if len(compressed) < len(s) else s
s = input("Enter a string: ").strip()
print("Compressed string:", compress_string(s))
```